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We present extensions to the cuRAMSES adaptive mesh refinement (AMR) code that enable cosmological hydrod

methods: numerical – cosmology: simulations – dark energy – neutrinos – dark matter

INTRODUCTION

The ΛCDM concordance model, with a cosmological constant Λ and cold, collisionless dark matter, provides an exc

Precision cosmological surveys such as DESI DESI2024, Euclid EuclidCollaboration2024, and the Vera C. Rubin Ob

In this paper we describe the implementation of three non-standard cosmological modules in cuRAMSES Kim2026

For the important case $c_s^2 = 0$, dark energy clusters at all scales identically: R_{DE} is independent of k. This allows t

$$\mathcal{S} = 1 + \frac{\Omega_{DE}(a)}{\Omega_{cb}} R_{DE}(k_{ref}, a),$$

evaluated at $k_{ref} = 10^{-3}\, h\,\mathrm{Mpc}^{-1}$ (any value suffices since R_{DE} is k-independent).

Quasi-Static Helmholtz Fallback

When no CAMB table is provided and $c_s^2 > 0$, a quasi-static approximation is available. In the sub-horizon, quasi-sta

$$\mathcal{F}_{DE}(k) = \frac{\tilde{k}^2 + \kappa^2}{\tilde{k}^2 + \kappa^2 + \alpha},$$

where $\tilde{k} = 2\pi\sqrt{k_x^2 + k_y^2 + k_z^2}$ is the dimensionless Fourier wavenumber, and

$$\kappa^2 = \frac{a^2\, E^2(a)}{c_s^2} \left(\frac{L_{box}}{c/H_0}\right)^2,$$

$$\alpha = \frac{3}{2}\,\Omega_\Lambda\, f_{DE}(a)\, \frac{1 + w(a)}{c_s^2} \left(\frac{L_{box}}{c/H_0}\right)^2.$$

For ΛCDM, $1 + w = 0$ so $\alpha = 0$ and $\mathcal{F}_{DE} = 1$. This approach is valid only for $c_s^2 > 0$; for $c_s^2 \to 0$, both κ^2 and α diverge

Namelist Configuration

The CPL dark energy module is activated by setting `de_perturb=.true.` in `&RUN_PARAMS`. The equation of state ar

```
&CPL_PARAMS
w0 = -0.9
wa = 0.1
cs2_de = 0.0
de_table = 'de_table.dat'
/
```

The code selects the solver path automatically: table-based if `de_table` is provided, quasi-static Helmholtz otherwise (w

MASSIVE NEUTRINO LINEAR RESPONSE

Physical Motivation

Massive neutrinos with total mass $\sum m_\nu > 0$ free-stream out of gravitational potential wells on scales below the ne

The standard approach of representing neutrinos as N-body particles Brandbyge2008,Viel2010 suffers from severe s

Transfer Function Table

We compute the neutrino response function

$$R_\nu(k, a) = \frac{T_\nu(k, a)}{T_{cb}(k, a)}$$

using CAMB, where T_ν and T_{cb} are the neutrino and CDM+baryon transfer functions, respectively. The table spans 200

The default cosmology uses Planck 2018 parameters PlanckCollaboration2020: $\Omega_b = 0.0486$, $\Omega_m = 0.3089$, $h = 0.67$

Poisson Solver Integration

FFT base-level solver

At the base level, where the grid is uniform and an FFT direct solve is performed, the neutrino correction is applie

$$\hat{\phi}(\mathbf{k}) = G(k)\, \hat{\delta}_{cb}(\mathbf{k}) \times \left[1 + \frac{\Omega_\nu}{\Omega_{cb}} R_\nu(k_{phys}, a)\right].$$

The physical wavenumber k_{phys} is computed via equation (**??**). This correction is applied in both the distributed (FFT

Multigrid fine-level solver

The MG solver operates in real space and cannot apply k-dependent corrections directly. Instead, the Poisson sour

$$4\pi G_{eff} = \frac{3}{2}\,\Omega_{cb}\, a\,\mathrm{scale},$$

where scale is the code-to-physical unit conversion. This correctly accounts for the fact that only CDM+baryon matter

Namelist Configuration

The neutrino module requires `use_neutrino=.true.` in `&RUN_PARAMS`, plus a `&NEUTRINO_PARAMS` block:

```
&NEUTRINO_PARAMS
omega_nu = 0.00140
neutrino_table = 'neutrino_table.dat'
/
```

At startup, the code validates that $\Omega_\nu > 0$ and the table file exists; otherwise the module is disabled with a warning.

Consistency with Dark Energy Corrections

When both neutrino and dark energy modules are active, the correction factors compose multiplicatively in equatio

SELF-INTERACTING DARK MATTER

Physical Motivation

In the CDM paradigm, dark matter is collisionless. Self-interacting dark matter (SIDM) models introduce a non-ze

Astrophysical constraints span several orders of magnitude in relative velocity: dwarf galaxies probe $v \sim 30$–$100\,\mathrm{km}$

Monte Carlo Scattering Algorithm

We implement a cell-based Monte Carlo scattering algorithm following the standard approach used in SPH/mesh c

Particle collection and pairing

We have presented three non-standard cosmological modules for the cuRAMSES AMR code:

CPL dark energy perturbations: a table-based linear response method using precomputed CAMB Weyl-potential rat

Massive neutrino linear response: scale-dependent Poisson equation corrections via precomputed $R_\nu(k, a)$ ratios, av

Self-interacting dark matter: cell-based Monte Carlo scattering with constant, velocity-dependent (Yukawa, power-l

All three modules are designed within a unified Poisson solver framework in which neutrino and dark energy correc

Together with the algorithmic optimizations described in Kim2026 — k-section domain decomposition, Morton key

Future work will apply these modules to constrained-realisation simulations targeting DESI and Euclid survey volu

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*Data Availability

The modified code is available at `https://github.com/kjhan0606/cuRAMSES-kjhan`. The CAMB table generation s

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COMPLETE NAMELIST REFERENCE

Table **??** lists all namelist parameters introduced by the non-standard cosmological modules.

Namelist parameters for non-standard cosmological modules.

Parameter	Namelist	Type	Default	Descripti
Global switches (&RUN_PARAMS)				
`sidm`	`&RUN_PARAMS`	logical	F	Enable S
`use_neutrino`	`&RUN_PARAMS`	logical	F	Enable n
`de_perturb`	`&RUN_PARAMS`	logical	F	Enable D
`dump_pk`	`&RUN_PARAMS`	logical	F	Dump m
CPL dark energy (&CPL_PARAMS)				
`w0`	`&CPL_PARAMS`	real	-1	Present-d
`wa`	`&CPL_PARAMS`	real	0	EOS time
`cs2_de`	`&CPL_PARAMS`	real	0	DE sound
`de_table`	`&CPL_PARAMS`	string	—	Path to c
Massive neutrinos (&NEUTRINO_PARAMS)				
`omega_nu`	`&NEUTRINO_PARAMS`	real	0	Ω_ν
`neutrino_table`	`&NEUTRINO_PARAMS`	string	—	Path to c
SIDM (&SIDM_PARAMS)				
See Table **??**				

CAMB TABLE FORMAT

Both the neutrino and dark energy response tables use the same ASCII format:

```
# Header comments (lines starting with #)
nk na
k_1 k_2 ... k_nk       [h/Mpc]
a_1 a_2 ... a_na
R(1,1) R(2,1) ... R(nk,1)
R(1,2) R(2,2) ... R(nk,2)
...
R(1,na) R(2,na) ... R(nk,na)
```

where $R(i, j) = R_X(k_i, a_j)$ is the response function. The k-values are in units of $h\,\mathrm{Mpc}^{-1}$ and the a-values are dimensio

The generation scripts `generate_neutrino_table.py` and `generate_de_table.py` (in `misc/`) produce tables in this